

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m;
```

```
    printf("Enter the number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the number of resources: ");
```

```
    scanf("%d", &m);
```

```
    int alloc[n][m], request[n][m], avail[m];
```

```
    int finish[n], safeSeq[n];
```

```
    printf("Enter the allocation matrix:\n");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("Process %d: ", i);
```

```
        for (int j = 0; j < m; j++)
```

```
            scanf("%d", &alloc[i][j]);
```

```
    }
```

```
    printf("Enter the request matrix:\n");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("Process %d: ", i);
```

```
        for (int j = 0; j < m; j++)
```

```
            scanf("%d", &request[i][j]);
```

```
    }
```

```
    printf("Enter the available resources: ");
```

```
    for (int j = 0; j < m; j++)
```

```
        scanf("%d", &avail[j]);
```

```
    for (int i = 0; i < n; i++)
```

```
        finish[i] = 0;
```

```

int count = 0;
while (count < n) {
    int found = 0;
    for (int i = 0; i < n; i++) {
        if (!finish[i]) {
            int ok = 1;
            for (int j = 0; j < m; j++)
                if (request[i][j] > avail[j]) { ok = 0; break; }
            if (ok) {
                for (int j = 0; j < m; j++)
                    avail[j] += alloc[i][j];
                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }
    if (!found) break;
}
if (count == n) {
    printf("System is in safe state.\nSafe Sequence is: ");
    for (int i = 0; i < n; i++)
        printf("P%d%s", safeSeq[i], i < n-1 ? " " : "\n");
} else {
    printf("Deadlock detected! Deadlocked processes: ");
    for (int i = 0; i < n; i++)
        if (!finish[i]) printf("P%d ", i);
    printf("\n");
}
return 0;
}

```

OUTPUT :

```
Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0
System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1
```